Decision-Driven Evidence Disclosure for Collaborative Security Correlation

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***Abstract*—Cross-organization detection can require corroboration from several local views, yet routine export of logs creates disclosure and operational costs. This paper presents a finite evidence protocol in which each organization retains raw events, emits a versioned signal, and decides whether to release one requested categorical aggregate. A coordinator requests the least costly untried predicate that can still affect an unresolved rule, counts independent provenance groups, and records supported, contradicted, or unresolved outcomes. An organization-local ledger commits each released representation and its charge atomically; replay of the same snapshot, recipient, and catalogue entry cannot produce another fine-grained release. We implement the protocol with a durable correlation worker and compare five sharing strategies over 1,000 deterministic protocol cases: five scenarios, ten seeds, four policy conditions, and five strategies. In the no-fault slice, decision-driven requests retained support for all 20 attack-labelled cases and rejected the 20 benign cases incorrectly supported by minimal sharing. Relative to fixed richer sharing, they used 540 rather than 660 policy units and sent 200,100 rather than 230,460 payload bytes. Denial, coarsening, and budget faults yielded unresolved cases rather than unsupported positive decisions. These are synthetic protocol measurements, not a field detection result or a formal privacy guarantee. A planned CIC-IDS2017 replay remains incomplete because the official archive endpoint returned a registration page.**

***Index Terms*—collaborative intrusion detection, selective disclosure, evidence provenance, policy accounting, security correlation**

I. Introduction

An organization can observe an authentication burst without knowing whether peers see the same pattern. A coordinator can improve the context of that alert by asking for independent observations. However, a request for raw login records would expose identities, timings, and internal topology that the partner may have no reason to disclose. Even a derived count can reveal a rare event. The design question is therefore about which evidence to request, when to stop, and how to account for repeated requests. NIST guidance frames participation, sharing rules, and information handling as explicit decisions rather than automatic publication [1]. Existing platforms make threat intelligence exchange practical [2]; controlled sharing work shows that an organization can choose how much to contribute [3]. Our narrower target is an incident-specific, executable disclosure protocol for finite alert predicates.

Consider an authentication anomaly. The initial signal says that a local detector saw a high-scoring authentication burst; it does not distinguish credential rejection from a provider outage. A coordinator may first ask for the coarse failure-rate category, then for failure kind, and stop when an outage alternative is established. A fixed rich profile would continue releasing the remaining available fields. In the implemented protocol, each response is either approved, coarsened to `unknown`, or denied. A refusal is a valid observation about policy, never evidence for the attack hypothesis. This matters because treating missing evidence as confirmation can amplify correlated false alarms.

We make three contributions. First, we specify a bounded, decision-driven request rule over a declared evidence catalogue and explicit terminal outcomes. Second, we implement local transaction-based release accounting, replay-safe responses, and provenance-aware correlation. Third, we evaluate the sharing and outcome trade-off under controlled faults with frozen cases and published output files. The contribution is a combination of semantics and implementation measurements. It does not claim a new anomaly detector, anonymous sharing, differential privacy, or real-organization detection accuracy.

II. Related Work and Boundary

MISP supports collaborative exchange of structured threat intelligence, including indicators and contextual events [2]. Its contribution is a mature sharing platform; it does not prescribe the specific sequence of incident predicates studied here. Freudiger *et al.* evaluate controlled data sharing for predictive blacklisting and show that collaborators can negotiate useful sharing with cryptographic tools [3]. Dara and Muralidhara identify privacy problems and cryptographic architectures for collaborative intrusion detection [4]. These works establish the feasibility and importance of selective sharing. Our protocol does not replace their cryptographic methods. It focuses on deterministic next-predicate choice, a bounded local catalogue, and explicit failure outcomes in an implemented correlation path.

Traditional intrusion-detection benchmarks address classifier performance under labelled traffic. CIC-IDS2017 was designed as a labelled network dataset with benign and attack periods [5], [6]. A replay against it could test detector and correlation quality, but our reported cases are generated by a local simulator, so benchmark accuracy cannot be inferred. Privacy claims also need care: Dwork and Roth formalize a guarantee in terms of neighbouring datasets and randomized mechanisms [7]. The integer units used in our ledger are policy

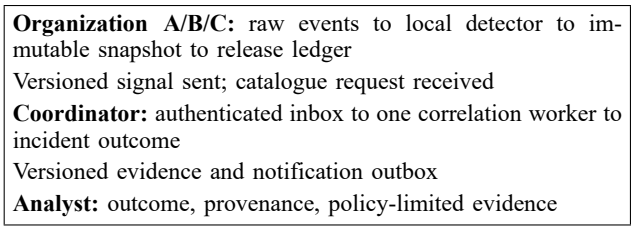


Fig. 1. Raw events stay in each organization. Only typed signals and approved catalogue responses cross the boundary.

costs set by an operator; they are not an ϵ value and do not establish that guarantee.

III. System and Threat Model

The system has n enrolled organization agents, one coordinator API, a durable correlation worker, and analyst clients. Each agent generates raw events and derived features locally. It is the authority for its snapshots and release policy. The coordinator receives a typed signal and may issue a request only for an entry in the finite catalogue. The authenticated recipient is `coordinator`; arbitrary SQL, log-file paths, and free-form queries have no wire representation. Figure 1 shows the trust boundary.

The adversary may submit repeated requests, delay or duplicate network delivery, correlate responses across time, or inject hostile content into local events. We assume enrolled keys are managed correctly, database storage on each agent is available during a transaction, and at most one representative vote is counted per enrolled provenance group. A malicious or compromised enrolled organization can still lie about a derived aggregate; the protocol does not attest to its underlying logs. Collusion between multiple nominally independent groups is outside the proven boundary. Correlation rules measure agreement with declared predicates, not the truth of a compromise.

A signal contains a unique signal and snapshot identifier, UTC observation interval, family, detector version and state, confidence or anomaly score, policy version, and recipient. The implementation limits observations to 1–600 seconds, accepts only supported schema versions, and rejects an unknown family before creating a correlation candidate. An incident groups signals by family and time bucket. The default bucket is 120 seconds, score threshold is 0.80, quorum is three independent groups, and request deadline is 90 seconds. These are configurable operational defaults, not tuned research parameters.

IV. Catalogue and Correlation Rules

The catalogue is intentionally small. Authentication has `auth_failure` (low/high/unknown; cost 1), `source_spread` (narrow/wide/unknown; cost 2), and `failure_kind` (rejected/unavailable/mixed/unknown; cost 2). Traffic has `request_rate` (normal/elevated/spike/unknown; cost 1), `source_spread`, and `service_errors` (low/high/unknown; cost 1). A coarsened response has value `unknown` and cost zero in the tested policy. The initial signal costs one unit. Values are enumerated, so a coordinator cannot request a raw IP address or an arbitrary finer range by varying a query string.

For a source o and family f, let V_o be its currently active released values. Its rule result is $d_f(V_o) \in \{S, C, U\}$ for support, contradiction, or unresolved. For authentication, S requires high failures, wide source spread, and rejected failures. Unavailable failures, low authentication failures, or narrow spread yield C. For traffic, S requires spike rate, wide spread, and high service errors; normal rate or low service errors yield C. Every other partial or unknown combination yields U. The short-circuit alternatives intentionally allow a decision without all three entries; they are rule semantics, not a probabilistic classifier.

An organization may submit multiple signals, but only one representative participates in an incident. Votes are then grouped by enrolled provenance identifier. If reports within a group disagree, its vote is unresolved. Given quorum q, the worker records S if at least q independent groups support, C if at least q contradict, and U otherwise; simultaneous support and contradiction quorums yield U with a conflict reason. The incident also records counts, pending requests, and coverage. Analyst workflow status (open, investigating, resolved) is separate from this evidential outcome. This prevents a case being marked resolved by a human from masquerading as proven technical support.

V. Decision-Driven Disclosure Protocol

The worker requests more evidence only while the incident is unresolved, has sufficient independent groups, and has not passed its deadline. For each unresolved participant it constructs the untried entries in the family catalogue. The implementation chooses

$$e^* = \operatorname*{arg\,min}_{e \in E_f \setminus (A_o \cup V_o)} \big(c(e), \mathrm{id}(e)\big), \tag{1}$$

where A_o is the attempted set, $c(e)$ is the declared unit cost, and the lexical identifier breaks ties. It asks for one entry and re-evaluates after the response. Selection is decision-driven in the limited sense that it runs only when the source rule remains unresolved and stops when the rule is decided. It is not an entropy optimizer or a proof of globally minimal disclosure. If every remaining predicate is blocked, the worker records a reason such as `no_useful_authorized_evidence`; it never fabricates a support vote.

Each request includes schema version, request ID, snapshot ID, catalogue entry, recipient, and UTC deadline. The agent checks the current policy, snapshot presence and age, and available budget within a database transaction. An approval stores the exact response and its charge under $(\text{snapshot}, \text{recipient}, \text{entry})$. A retry for that tuple reuses the stored release and does not debit again. If policy has since tightened from fine to coarse, the old fine response is denied for further delivery. A denial carries no evidence value or identifier and has zero cost. Transaction failure rolls back both response and debit.

There are two hard limits: a per-snapshot cap and an organization-local rolling-hour cap. Consequently, creating more request IDs for one snapshot does not gain new values, while opening new snapshots still consumes the rolling cap. The initial signal is accounted before entering the durable agent outbox. The outbox retries transient failures with backoff and stable jitter; permanent rejection increments a visible gap counter. These rules handle ordinary retries and bounded release, but they do not prevent an attacker who controls many enrolled organizations or arbitrarily many time-separated snapshots from gathering a large cumulative view. A deployment must govern enrollment, retention, and policy epochs as well as numerical caps.

VI. Reference Implementation

The implementation uses a FastAPI API, PostgreSQL incident and evidence tables, a single worker that owns correlation transitions, and SQLite WAL ledgers in organization agents. The API validates typed Pydantic contracts and authenticates node keys before accepting signals or evidence. Unique identifiers make inbox processing and participation idempotent. The worker processes bounded batches, locks candidate incidents, updates their outcome, and writes notification records in the same database transaction. It also revisits pending incidents for deadline and evidence expiry, so an absence of new traffic does not leave an expired request active indefinitely.

Evidence records have a version and active flag. A correction or withdrawal changes the incident revision and can invalidate derived explanation state. The analyst view receives versioned notifications and can recover from a cursor if a bounded live queue overflows. PostgreSQL migrations and backup/restore were exercised locally; these checks establish that the tested schema can start and recover in that environment, not that a multi-site disaster recovery objective has been met. The correlation worker is deliberately singular in the reference deployment, limiting concurrent state ownership. It is a throughput and availability limit to revisit only with a measured need.

VII. Evaluation Design

We froze `research/cases.json` and its SHA-256 digest before recording outputs. The five scenarios are normal traffic, authentication attack, provider outage, traffic attack, and flash crowd. Each is run with ten seeds, three synthetic nodes, and the same 10-second event window. Attack ground truth consists of the two attack scenarios, giving 20 positive and 30 benign cases per strategy–fault slice. The local generator varies seeded event content but does not create independent real organizations; case counts are protocol repetitions, not a sample of deployments. Ground-truth labels are used by the evaluator and do not enter wire messages.

We compare (i) local only, which accepts the local high-score alert; (ii) fixed minimal, which transmits the initial signal but requests no aggregate; (iii) fixed richer, which attempts catalogue entries in fixed order; (iv) static escalation, which attempts that order in separate rounds; and (v) decision driven, which uses Eq. 1 and stops after a rule decision. Four conditions are run: no fault, policy denial of `failure_kind`/`service_errors`, coarsening of `source_spread`, and an exhausted per-snapshot budget of two units. The full factorial design is $5 \times 10 \times 5 \times 4 = 1{,}000$ cases. Each strategy–fault cell has 50 cases; the outcome calculation in the harness requires all three node decisions to agree, matching the configured quorum but without exercising arbitrary provenance conflicts.

Primary measurements are positive attack cases supported (TP), benign cases supported (FP), unresolved cases, policy units, released refinements, serialized payload bytes, and local protocol time. Payload bytes sum the encoded signal, request, and response bodies; they exclude TCP, TLS, database, and deployment overhead. Time is local wall-clock execution on one Windows CPU machine and excludes network transit. The disclosed integer cost is an accounting metric, not a privacy loss measure.

TABLE I
No-fault synthetic slice; 50 cases per strategy.

Strategy	TP	FP	Units	Releases	Bytes
Local only	20	20	0	0	0
Fixed minimal	20	20	120	0	48,180
Fixed richer	20	0	660	360	230,460
Static escalation	20	0	660	360	230,460
Decision driven	20	0	540	300	200,100

TABLE II
Decision-driven strategy under injected policy faults.

Condition	TP	FP	U	Units	Bytes
None	20	0	0	540	200,100
Deny	0	0	40	240	164,640
Coarsen	0	0	20	420	200,340
Budget	0	0	40	240	164,640

VIII. Results

Table I shows the no-fault slice. Local-only and fixed-minimal retain 20 true positives but also label 20 benign provider-outage or flash-crowd cases as supported. Richer evidence eliminates those false positives under the declared rules. The decision-driven strategy reaches the same 20 supported attack and zero supported benign cases as fixed richer and static escalation, while using 120 fewer units (18.2%), 60 fewer refinements (16.7%), and 30,360 fewer payload bytes (13.2%) than fixed richer. These are aggregate differences across the 50 fixed cases, not uncertainty-bounded estimates of population performance.

Table II makes the safety cost visible. Under denied evidence and exhausted budget, 40 of 50 decision-driven cases are unresolved; under coarsening, 20 are unresolved. In all three conditions its positive count is zero and its false-positive count is zero. “No false positives” here should not be read as good detection under faults: attack decisions are also withheld. A policy owner can therefore limit exposure, but analysts must see the unresolved workload rather than silently treating it as benign.

The no-fault mean round count is 2.0 for decision driven, 0.8 for fixed richer, and 2.4 for static escalation. This measure records the harness request schedule, not end-to-end user latency. The respective median local protocol times are 0.353, 0.391, and 0.379 seconds, but process scheduling and file-system variability were not controlled tightly enough to claim a latency improvement. The expected deployment trade-off is fewer disclosed bytes versus potentially more request-response round trips.

IX. Worked Trace and Cost Decomposition

The result is easier to interpret by tracing one source through each family. An authentication-attack window produces a high local score and an authentication-burst signal. The agent's private snapshot has `auth_failure=high`, `failure_kind=rejected`, and `source_spread=wide`. The coordinator's first request is `auth_failure`, which costs one unit. The source remains unresolved because high failure volume can be an outage. The next request is `failure_kind`, chosen before equally priced `source_spread` by lexical order. A rejected value still leaves the source unresolved until wide spread is known. The third approval completes the source's supporting rule. The initial signal plus three refinements cost $1+1+2+2=6$ units. All three independent synthetic groups must complete that path for the incident to be supported.

For a provider outage, the first request also returns high authentication failure, but the second returns `failure_kind=unavailable`. That value contradicts the attack rule immediately. The decision-driven strategy does not request `source_spread`; its per-source charge is $1+1+2=4$ units. Fixed richer still requests the third entry and reaches the same contradiction after spending six units. This example illustrates the paper's specific notion of decision value: an entry is worth requesting only while the declared rule needs it. No attempt is made to infer a private value from the owner refusing to release it.

The traffic family has an analogous path. An attack source reports `request_rate=spike`, `service_errors=high`, and `source_spread=wide`. The first two requests cost one unit each, and the final spread request costs two, giving five units with the initial signal. A flash crowd has spike rate but low service errors; it is contradicted after the second request at a total of three units. Fixed richer proceeds to the spread entry and pays five. In the ten normal cases, no high-scoring family candidate is issued, so neither the initial signal nor a refinement is charged. The no-fault totals follow directly from these four abnormal scenario paths, each repeated for ten seeds and three agents: decision driven uses $30(6+4+5+3)=540$ units; fixed richer uses $30(6+6+5+5)=660$. This arithmetic is a check on the summary CSV, not a claim that the chosen catalogue costs measure real information exposure.

This controlled decomposition also exposes the narrowness of the metric. A two-unit `source_spread` category might be less sensitive in one real organization than a one-unit `failure_kind` category in another. The costs are operator-specified labels and were held fixed only to make strategy comparison possible. The legitimate operational interpretation is that the same rule outcome was reached with fewer approved catalogue entries under this fixed policy. A privacy interpretation would require measuring inference risk about individual events or organizations under the actual query history. A bandwidth interpretation would require the full transport stack, which the byte metric intentionally omits.

TABLE III
No-fault trace per active synthetic source. The normal scenario emits no candidate.

Scenario	Final rule	Decision units	Richer units
Authentication attack	Support	6	6
Provider outage	Contradict	4	6
Traffic attack	Support	5	5
Flash crowd	Contradict	3	5

X. Correctness Properties and Open Risks

The design treats inability to decide as a first-class result. An evidence request may be denied by policy, budget, expiry, or a missing snapshot. The response then has no evidence identifier and no value. Coarsening yields an explicit `unknown` category. Neither can satisfy a positive predicate. This is why denial and exhausted-budget faults suppress both false positives and true positives in Table II. The system still records the original local anomaly and the reason the cross-organization rule remained unresolved. An analyst can investigate locally, but the coordinator must not rewrite that case as benign merely because a partner protected its data.

Idempotency exists at several boundaries. A snapshot identifier can be queued once, and a conflicting reuse of that identifier is an error. An agent stores a released response by snapshot, recipient, and entry, so a fresh network request ID for the same data does not create a second release. A coordinator records signals in an inbox and processes a bounded batch through one worker. Participation is unique for the incident and organization. The release ledger and correlation database are distinct systems; there is no distributed transaction across them. If an owner commits a release but the coordinator never receives the response, a retry retrieves the stored representation without another charge. If an incident times out before that retry arrives, the release has still happened and its cost remains in the owner ledger. That is an honest accounting of exposure even when it brought no analytic benefit.

The rule is deterministic but not necessarily robust to adversarially crafted data. An attacker who drives a legitimate source to show `failure_kind=unavailable` can force a contradiction in the authentication family, and an attacker who suppresses service errors can force a contradiction in the traffic family. The reference detector and categorical thresholds were not trained or validated against evasion. We therefore use *supported* and *contradicted* as names for rule states rather than "attack confirmed" and "safe." The analyst-facing evidence record should include each predicate and provenance group so that a responder can review the basis of a decision. A stronger

TABLE IV
NO-FAULT SERIALIZED PAYLOAD BYTES BY SCENARIO, TEN CASES EACH.

Scenario	Decision	Richer	Saved
Normal	0	0	0
Authentication attack	57,690	57,690	0
Provider outage	42,600	57,780	15,180
Traffic attack	57,510	57,510	0
Flash crowd	42,300	57,480	15,180
Total	200,100	230,460	30,360

deployment would add source integrity controls, calibration, and independent adversarial testing.

The worker also has an operational liveness limit. It creates one request per unresolved participant when there is no pending request, re-evaluates after responses, and stops at a deadline. A slow participant can leave the whole three-vote investigation unresolved. Increasing the deadline improves the chance of obtaining all responses but extends the time before a final outcome. Reducing it improves responsiveness but raises unresolved workload. The chosen 90-second default was not optimized in the synthetic harness, whose local protocol timings are far shorter. A deployment should measure response-delay distributions and choose a deadline based on analyst workflow needs. The correct timeout policy is a service-level decision, not a property of the greedy selection rule.

Finally, the trust boundary is not a secrecy proof. An approved category can reveal that an organization is observing abnormal traffic; a sequence of categories can reveal trends. The coordinator also sees which owner denied a request and when it did so. Authenticated transport and local ledgers restrict access and produce an audit trail, but they do not eliminate inference. We recommend treating these fields as sensitive operational data with access controls and retention limits. The reported budget cap is an engineering control for that surface and should be revised with a domain-specific risk assessment.

XI. Scenario-Level Measurement Check

Table IV breaks the aggregate byte result into the five frozen scenarios. There is no saving in either attack scenario because every positive predicate must be seen before support is possible under the declared rule. All savings occur in the two benign alternatives that can be contradicted early. For provider outage, decision driven sends 42,600 serialized bytes across ten cases versus 57,780 for fixed richer. For flash crowd it sends 42,300 versus 57,480. The two differences sum to the reported 30,360 bytes. This distribution matters: a deployment with few short-circuit benign cases would see much less saving, even if its total event volume were higher. Conversely, a catalogue with more expensive predicates following common benign alternatives might see more saving. Our result must not be extrapolated without a deployment's event mix and policy costs.

The same table also helps audit what the byte metric does and does not count. The initial signal is serialized once per active synthetic source. Each attempted refinement adds a request body and a response body, including denials and coarsenings. Byte counts vary slightly between semantically similar cases because identifiers and JSON values have different lengths. The measurement does not count raw event size, since raw events never cross the organization boundary in this protocol. It also does not include HTTP headers, TLS records, retries, or notifications sent to an analyst. The saving from omitting a refinement is therefore a saving in the application message bodies generated by this particular harness. Operational bandwidth should be remeasured at the transport boundary.

The synthetic cases were designed to test an ambiguity that a local high-score alert cannot settle. In a provider outage, a failed login may return a server error rather than a credential rejection. In a flash crowd, request volume rises without the service-error condition required by the traffic rule. The local-only and fixed-minimal strategies have no distinguishing evidence, so the harness labels the high-score anomaly as supported; this creates 20 false positives relative to the generator labels. The richer strategies ask enough questions to reject those benign alternatives. A different anomaly detector might distinguish them from local data, reducing the value of collaboration. The experiment holds the detector constant because it asks about the disclosure protocol; it does not prove that collaboration is better than improving local detection.

Two kinds of validity are distinct here. *Internal protocol validity* asks whether the code obeys the specified catalogue, budget, idempotence, and stopping rules. The frozen case manifest, direct CSV rows, sum checks, and recovery tests provide evidence for that claim. *External detection validity* asks whether the chosen predicates, costs, thresholds, and provenance assumptions generalize to realistic network traffic and organizations. This study cannot answer that question. The 50 cases in one no-fault slice are five authored scenarios under ten seeds, and the same generator supplies all three synthetic nodes. Reporting a conventional confidence interval over those 50 rows would overstate independence. We therefore report exact finite-case totals and mark wider conclusions as hypotheses for later study.

The failure matrix is similarly a protocol stress test. Denial is injected for a specified pair of entries; budget exhaustion is implemented by reducing the per-snapshot cap; coarsening forces one entry to `unknown`. These are controlled interventions, not observed rates of real policy actions. The outcome semantics can be checked under them: a denied or unknown predicate does not become positive evidence, and a case unable to meet a rule remains unresolved. But the intervention cannot show how human analysts handle those cases, whether a partner would refuse more often for specific incidents, or how much information the coordinator could infer from refusal itself. Those require a field or participant study with explicit consent and access governance.

XII. Operational Semantics and Invariants

The implementation can be described as a transition system. Let I be an incident state, S_o an organization snapshot, $R_{o,e}$ a request, and L_o the organization-local release ledger. The

TABLE V
PROTOCOL TRANSITIONS AND DURABLE EFFECTS.

Event	Guard	Durable effect
Signal	Valid schema, enrolled node, score threshold	Inbox row and one organization participation
Request	Unresolved, live, no pending source request	One typed catalogue request
Approve	Policy and both caps permit	Exact response and charge in one ledger transaction
Coarsen	Owner policy requires coarse value	`unknown` representation; zero refinement charge
Deny	Policy, missing snapshot, expiry, or budget	No evidence; source cannot be silently counted
Expire	Deadline or evidence TTL reached	Re-evaluated incident and versioned notification

coordinator creates a request only if I is unresolved, its deadline is live, source o has no pending request, and e belongs to the family's catalogue. The owner responds with one of $\{\text{approved}, \text{coarsened}, \text{denied}\}$. The worker validates that the response matches the pending request, persists any evidence as a new version, and re-evaluates I. A terminal outcome cancels outstanding requests. A denied or expired request prevents further refinement for that source in the current investigation. This is intentionally conservative: a later policy change needs a new investigation or an explicit restart operation rather than silent indefinite probing.

The release invariant is that, for any snapshot s, recipient r, and entry e, at most one stored representation exists: $|L_o[s, r, e]| \leq 1$. The charge invariant is $\text{charge}(s, r, e) = 0$ for a denied or coarsened response and the catalogue cost for an approved response. The transaction writes both response and charge, so a crash before commit exposes neither, and a crash after commit can replay the exact representation. The per-snapshot and rolling-hour inequalities are

$$\sum_{x \in L_o(s)} c_x \leq B_s, \qquad \sum_{x \in L_o[t-3600, t]} c_x \leq B_h. \tag{2}$$

The second sum spans snapshots owned by one agent. It is neither a distributed global budget nor an ϵ composition theorem. It makes repeated local requests visible and bounded within one window. Responses are immutable at their release version; a tightened policy refuses reuse of an earlier fine response, while already received information cannot be retroactively erased from a recipient.

The outcome invariant is that only independent provenance groups contribute votes, and no `unknown`, denial, timeout, missing snapshot, or absent response can count as support. A single organization's multiple signals remain one participation for the incident. Conflicting votes within a provenance group reduce that group to unresolved. If both supporting and contradicting quorum conditions hold, the incident is unresolved with a conflict reason. This conservative combination protects against several straightforward counting errors, although a forged provenance assignment or colluding organizations can defeat independence assumptions.

For each incident revision:

1) Recompute active evidence, provenance votes, and outcome.
2) If supported or contradicted, cancel pending requests and publish the revision.
3) If deadline passed, publish unresolved with deadline reason.
4) For each unresolved source without a pending or denied request, choose the cheapest untried catalogue entry.
5) If no useful entry remains, publish unresolved with coverage and reason.
6) Persist one request; on response or expiry, start again at step 1.

Fig. 2. Coordinator transition loop. Every transition is persisted before a notification is offered.

A. Why the greedy choice is bounded but not optimal

Equation 1 makes a reproducible local choice and avoids an adaptive search over raw data. It can request a cheap entry that ultimately proves unnecessary. For example, an unavailable authentication failure kind alone contradicts the attack rule, but `auth_failure` has lower unit cost and is requested first. A clairvoyant planner could ask the decisive kind immediately, but it would need a prior on local values and could favor fields with higher disclosure risk. The current rule avoids speculative utility estimates. It is best described as a deterministic adaptive baseline with rule-based stopping, not an optimal active-learning policy.

B. Failure boundaries

Network retries can duplicate delivery; unique signal and request identities, stored response representations, and transactional worker state make repeated processing operationally safe. They do not solve every distributed failure. A partition that lasts past the deadline produces unresolved rather than an inferred benign result. An agent that loses its SQLite file may lose release history; backups and recovery need an operator procedure before restarting that identity. A coordinator can learn from approved aggregates even if it later deletes its database, so retention and access rules remain necessary. The design also lacks a cryptographic commitment from raw logs to aggregate values; fabricated aggregates must be addressed by governance or stronger attestation in a later system.

C. A bounded coordinator loop

Figure 2 makes the request sequence explicit. It is a bounded loop because the catalogue has three entries per family, each attempted entry is recorded, and the incident has a deadline. The worker does not keep asking a source after a denial or expiry. An approved but nondecisive response can lead to another request only after the previous request has been applied to the incident state. This matters for backpressure: an agent never needs to handle an unbounded fan-out of speculative queries for the same incident. At most one request per participant is pending at a time in the reference implementation.

An owner has its own bounded loop. It first parses the exact request schema and rejects unknown catalogue entries or recipients. It checks the deadline and current policy, then looks up the immutable snapshot. If a release for the tuple already

exists, it returns the stored response unless the new policy forbids repeating the earlier fine value. Otherwise, it computes the permitted representation and prospective charge, checks the snapshot and rolling caps, inserts the response and charge, and commits. Network I/O occurs after the transaction. This ordering is essential: sending a response before committing a charge would permit an unrecorded release after a crash; charging before storing a response would create a debit with no replayable representation. The SQLite transaction is local to the owner and can therefore enforce the invariant without a distributed commit protocol.

The coordinator cannot prove that an owner actually enforced its ledger. It can validate the response contract, policy version, request linkage, and evidence identifier, but a compromised agent could lie about its local accounting or values. This is a distinct limitation from coordinator misuse. A stronger trust model might require remote attestation, signed release receipts, or an independently audited hardware module. Those mechanisms add deployment complexity and are not implemented here. The design instead makes the owner boundary clear and allows an organization to examine its own durable ledger. A complete security analysis should test key rotation, compromised enrolment credentials, replay outside the incident window, and malicious provenance claims.

The release tuple is coarser than the network request ID. Two coordinators asking for the same entry against the same snapshot and recipient see the same committed representation, not two independently sampled answers. This simplifies both audit and retry semantics but prevents the owner from changing a value within one snapshot without issuing a new snapshot identity. That is appropriate when a snapshot is immutable. If source data must be corrected, a later revision should be represented explicitly and prior evidence should become inactive in the coordinator. Reusing a snapshot ID with different underlying values would make the ledger's idempotence guarantee ambiguous and is rejected by the implementation.

D. Policy changes during an investigation

An investigation may outlive a policy update. Suppose an organization has approved `auth_failure=high` and then tightens its policy before the coordinator asks for `failure_kind`. The second request is evaluated under the new policy version. If the new policy denies that entry, the response is a denial; previously approved evidence remains part of the coordinator's incident record until it expires or is explicitly withdrawn. The correlation rule cannot complete from the remaining values and records unresolved. This differs from pretending the earlier approval never happened. A correct audit must retain both facts: data was released under the old policy and later disclosure was refused under the new one.

Another case is a retry of an earlier approved entry after the policy now requires coarsening. The ledger refuses to resend the earlier fine value. It also does not synthesize a new coarse value under the same release tuple, because that would create two representations for one immutable snapshot entry and complicate the at-most-once interpretation. A future version could support explicit supersession with a monotone representation order, but then it would need to prove that a later response never reveals more than the earlier one and to account for both transmissions. The present rule is conservative and simple to audit. It sacrifices some liveness when policy changes mid-incident.

Budget exhaustion has a comparable race. Two requests for different entries can arrive concurrently. A read-then-write check outside a transaction could allow both to observe remaining capacity and jointly exceed the cap. The owner uses a write transaction around cap checking and release insertion, serializing those decisions. The first eligible request can commit; the second sees the updated sum and may be denied. This produces order dependence when the budget cannot cover both entries, but it does not overspend. The coordinator's one-pending-request-per-source rule makes that race rare in the reference flow; the ledger still enforces the cap if callers bypass that scheduling expectation or if old retries arrive late.

E. Evidence freshness and incident revisions

The worker's outcome is a function of currently active evidence, not an irreversible label. Evidence has an age limit and can be corrected or withdrawn. When an item becomes inactive, the worker recomputes source decisions and provenance votes, then increments the incident revision if the outcome, reason, or summary changes. A formerly supported incident may become unresolved when a required predicate expires. This is the safer direction for stale evidence: it does not preserve a positive conclusion whose basis is gone. A downstream explanation generated from the older revision must also be marked stale, which is handled by the separate explanation lifecycle described in Paper 2. The sharing protocol here does not rely on an LLM to determine the outcome.

Freshness checks themselves can fail operationally. If the worker is stopped, an expiry may not be reflected in the materialized incident immediately, even though the underlying evidence timestamp is old. The implementation revisits expired evidence when the worker runs again and updates the incident. A high-assurance reader should also check evidence validity at read time, or expose a "worker delayed" health state, before acting on the last materialized outcome. The tested local deployment verifies recovery behavior but does not demonstrate a hard upper bound on stale exposure under prolonged worker outage. That bound would need monitoring, alerting, and an availability objective.

Provenance requires similar lifecycle discipline. A group identifier is meaningful only while it refers to an enrolled and active source. If an organization is disabled, its evidence should not continue contributing a fresh vote. The worker's query joins only active organizations and recomputes outcomes. This guards against one class of stale membership error. It does not establish that a group truly corresponds to an independent underlying sensor. A partner could relay the

same upstream signal through several identities, or a shared cloud provider could make ostensibly separate organizations fail together. In such settings, quorum should be defined over documented failure domains, not just account IDs. Establishing those domains is a governance task that the protocol makes explicit but cannot automate.

F. A useful boundary on formal claims

The invariants above are operational properties of the implemented state transitions. They are checked by code paths and targeted tests; this manuscript does not provide a machine-checked proof over all interleavings. In particular, the owner ledger's transaction and the coordinator database's transaction are separate. A message can be lost between them. The system is designed so such loss spends the owner's budget and may leave the incident unresolved, rather than allowing an uncharged release or fabricating a decision. A formal model could express this as a safety property under asynchronous delivery with eventual retries, then verify liveness only under a bounded-delay assumption. The present evaluation exercises representative failures but stops short of that formal verification claim.

XIII. Limitations and Threats to Validity

The scenarios are hand-authored and use a simplified local detector. Their benign alternatives are known to the authors and may favor the rule. Ten seeds vary event realizations but do not create independent traffic populations. The outcome metrics count support under declared predicates, not verified compromise. The synthetic three-node harness evaluates the protocol functions and local ledgers; it is not a WAN deployment, a privacy attack study, or a measurement of multi-tenant service capacity. Network transit, TLS framing, and analyst decision quality were not measured. The no-fault reduction is therefore a reproducible local comparison within a finite catalogue, not a universal savings rate. Refusal and exhaustion substantially increase unresolved cases, showing that lower disclosure can have an operational cost. The official CIC-IDS2017 replay is uncompleted, and primary references should be revalidated against final venue requirements before submission.

XIV. Conclusion

Decision-driven disclosure gives a coordinator a bounded way to ask only while a declared rule remains unresolved and gives the owner local authority over each answer. In the controlled no-fault evaluation it preserved the richer baseline's rule outcomes with lower policy cost and fewer serialized bytes; injected policy restrictions converted many cases to unresolved. This supports a concrete protocol and implementation result. Field detection, privacy leakage, and human usefulness require separate evaluation.

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